32-bit microcontroller

How to improve ADC sampling accuracy

Application Note

Rev1.0 November 2023

Applicable objects

Product series	Product model
HC32A/F/L/M	All models

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1 Overview

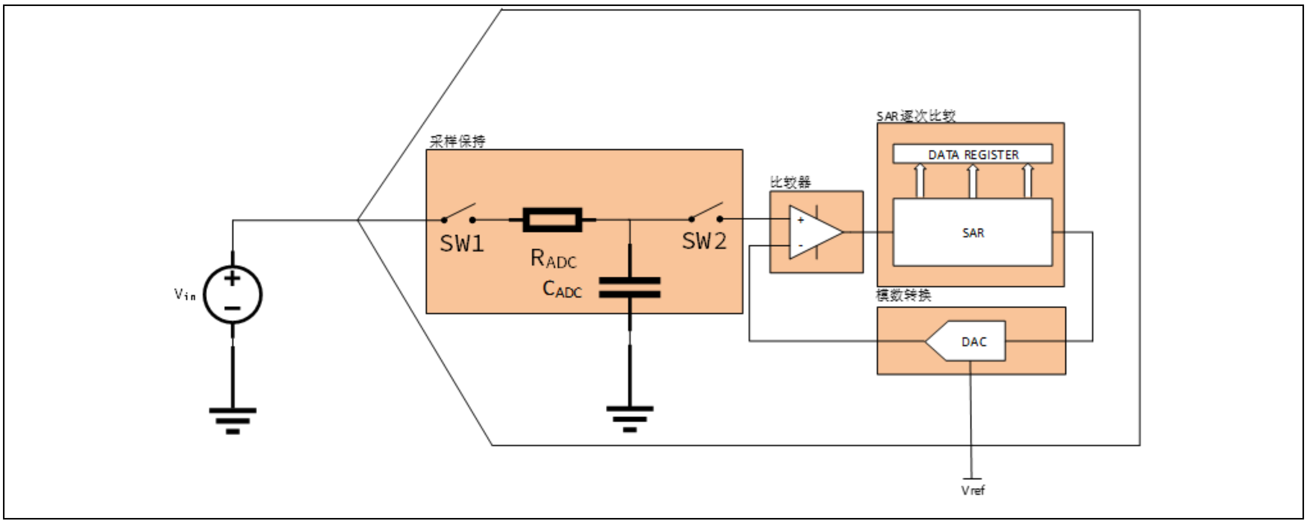
HC32A/F/L/M series products are MCUs launched by Xiaohua Semiconductor based on the ARM cortex core. Each MCU has 1~3 successive approximation digital-to-analog converters (SAR ADCs) built in. Each unit can support 8~16 external pin input channels or specific internal voltage, temperature and other signal sampling. The highest accuracy is 12 bits, the highest sampling conversion rate can reach 2.5 mbps, the internal channels are flexibly mapped, the sampling results can be read through DMA, and can be linked with the multi-function TIMER to achieve precise control of the sampling timing. It can meet the needs of most consumer, home appliances and industrial products for ADC.

This article describes the principle and inherent error of SAR ADC, the main factors affecting its sampling accuracy, and how to improve the sampling accuracy of ADC.

2 Introduction to SAR ADC

2.1 SAR ADC principle

Successive approximation ADC (SAR ADC), as the name implies, converts analog signals into digital signals by comparing the input signal with the reference source voltage step by step. The conversion process of SAR ADC is divided into two stages: sampling stage and conversion stage.



Sampling stage:

SW1 is closed, SW2 is disconnected, and Vin charges the sampling capacitor CADC through the internal switch RADC. After the sampling cycle ends, it enters the conversion stage.

Conversion stage:

SW1 is disconnected, SW2 is closed, and the voltage on the sampling capacitor (ideally Vin) is compared with (1/2) * V_{REF}. If Vin ≥ (1/2) * V_{REF}, the highest position is set to 1, and the DAC inputs the next comparison value (3/4)* V_{REF}. Vin is then compared with (3/4) * V_{REF} to determine the value of the second highest position. According to the comparison result, the DAC value is output again, compared, and finally the conversion value is obtained.

2.2 SAR ADC inherent error

The principle and structure of SAR ADC determine that it has several inherent errors.

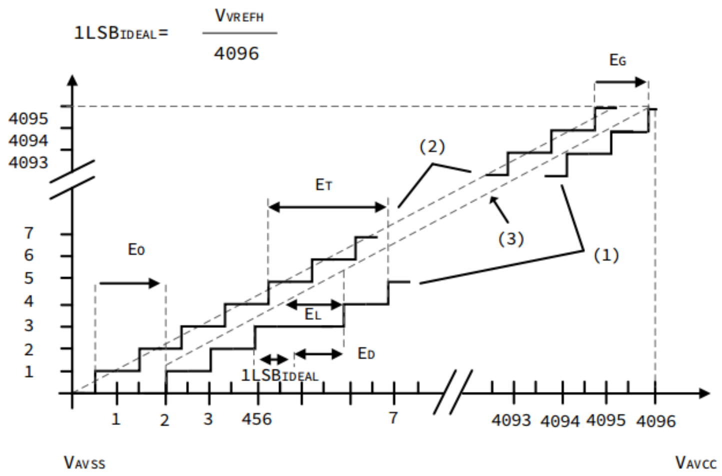


Figure 2-1 SAR ADC error

- Offset error E_O

  Offset error refers to the difference between the actual input analog quantity and the theoretical input analog quantity when the conversion value is 1. The offset error can be positive or negative.

- Differential nonlinearity error E_D

  Differential nonlinearity error refers to the maximum difference between the actual analog quantity step and 1 LSB when the conversion value changes by 1.

  $$E_D = \text{actual analog change - 1 LSB}$$

- Integral nonlinearity error E_L

  Integral nonlinearity error refers to the maximum deviation between the actual analog quantity and the actual conversion curve.

- Total unadjusted error E_T

  Total unadjusted error refers to the maximum error between the actual analog quantity and the ideal conversion curve.

- Gain error E_D

  Gain error refers to the error between the actual analog quantity and the theoretical analog quantity in the last conversion.

3 ADC external error sources and solutions

3.1 Reference source

The principle of SAR ADC is to obtain the conversion result by comparing the input signal with the reference voltage, that is:

$$Code_{ADC} = \frac{V_{in}}{V_{ref}} * 2^N \qquad (1)$$

From equation (1), it can be seen that V_{ref} has a great influence on the accuracy of ADC, which is mainly manifested in the following aspects:

3.1.1 Reference source voltage range

The reference source voltage needs to meet the following requirements:

1. It should be higher than the highest voltage of the input signal and lower than or equal to the ADC power supply voltage.
2. It should be as close to the input signal range as possible.

- The reference source voltage should be higher than the input signal voltage to accurately measure the extreme value of the input signal. It should be lower than or equal to the ADC power supply voltage to prevent the reference voltage source from backflowing the ADC power supply.
- The reference source voltage range should be close to the input signal range to maximize the theoretical accuracy. For example:

For a 12-bit ADC, the input signal range is 0~1.8 V.

If the reference power supply voltage is selected as 3.3 V, 1 LSB= $3.3/2^{12}$=3.3/4096 =0.81 mV. In this case, a theoretical accuracy of 0.81 mV can be achieved.

If the reference power supply voltage is selected as 2.0 V, 1 LSB= $2.0/2^{12}$=2.0/4096=0.49 mV. In this case, a theoretical accuracy of 0.49 mV can be achieved.

Optimization suggestions:

- A reasonable reference source voltage range can significantly improve the theoretical sampling accuracy. The reference source is slightly larger than the maximum amplitude of the input signal. Of course, in actual product applications, the reference source is not necessarily selected arbitrarily. It may be the power supply voltage. Then, the input signal is appropriately amplified to make its maximum amplitude close to the reference source range, which can also reduce the accuracy loss. For Xiaohua Semiconductor MCU, in addition to being slightly larger than the input signal, the reference source also needs to meet the special requirements in the specification, such as: Xiaohua Semiconductor's HC32F4A0 requirements.

Symbol	Name	Minimum	Typical	Maximum
V_{ref}	Positive reference voltage	1.8 V	-	AVCC

$0 \le V_{ref}$-AVCC≤ 1.2 V

3.1.2 Reference source accuracy

It is not difficult to understand from equation (1) that the conversion value of ADC is the relative value of the input signal and the reference source. Only when the reference source has high accuracy can the output result of ADC be accurate. The accuracy of the reference source is affected by factors including temperature coefficient, linear regulation rate, load regulation rate, etc. These factors need to be paid attention to, especially in high-precision occasions.

If the input signal of 12-bit ADC is 2.5 V and the reference voltage is 3.3 V, the converted value is:

$$\frac{2.5}{3.3}*2^{12}=0xC1F$$

If the reference source accuracy is 1%, assuming 3.3*0.99 = 3.267 V, the converted value is:

$$\frac{2.5}{3.267}*2^{12}=0xC3E$$

$$Code_{\Delta}=0xC3E\text{-}0xC1F=31\ LSB$$

So 1% accuracy is far from meeting the 1 LSB accuracy requirement, and the accuracy needs to be improved.

Optimization suggestions:

- In high-precision situations, it is often necessary to use a high-precision voltage reference IC to provide a reference source for the ADC.

3.1.3 Reference source noise

During the ADC conversion process, the switching and charging process of the internal capacitors of the ADC will cause input spike currents. These spike currents will cause spikes in the reference source voltage, thereby affecting the sampling accuracy of the ADC, especially when the input signal is large. The reference source noise has a more obvious impact. Therefore, it is very important to reduce the noise of the reference source voltage.

Optimization suggestions:

- On the one hand, you can choose a low-noise LDO or voltage reference as the ADC reference source.
- On the other hand, Xiaohua Semiconductor also recommends that customers configure low-ESR ceramic capacitors near the chip reference source for noise suppression, and provide nearby energy storage for the reference source. Taking HC32F4A0 as an example, its data sheet requires the following configuration:

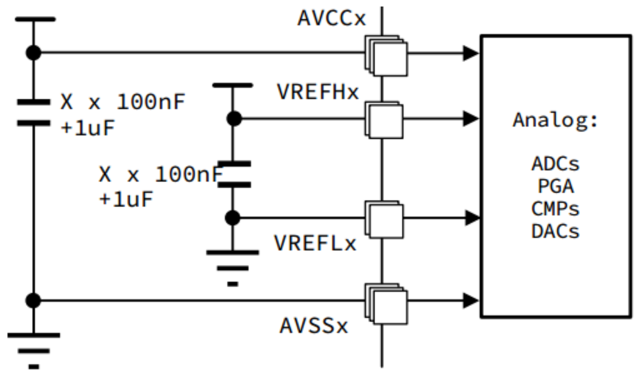


Figure 3-1 HC32F4A0 ADC reference source capacitor

- During PCB layout, ensure that the reference power supply loop is as small as possible and separated from the digital ground, using single-point grounding and using the same ground plane as the analog power supply ground. Energy storage and decoupling capacitors should be as close to the power pins as possible.

3.2 Power supply noise

The influence of reference source noise on ADC accuracy has been discussed above. In fact, a "clean" ADC power supply is also crucial for ADC. Power supply noise may affect the internal circuit of ADC, or it may be transmitted to the reference source or input signal, resulting in sampling error.

DC-DC power supply: DC-DC power supply transfers energy in the form of switching. There are high-frequency switching signals in its topology, accompanied by charging and discharging of inductors and output capacitors. Switching signals may induce spikes at the output of the power supply or the ground plane, and the charging and discharging process of the inductor and output capacitor will cause the output voltage to have obvious ripples. These can be regarded as noise from the power supply.

Linear LDO: Linear LDO is based on linear loop control, has no switching signals, does not generate spikes, and has small ripples. LDOs with high power supply rejection ratio (PSRR) and low noise are ideal analog power supply solutions.

Optimization suggestions:

- For power supplies for analog modules, it is best to use LDOs with high PSRR and low noise to avoid directly using switching power supplies to introduce large spikes and ripples.
- Similarly, Xiaohua Semiconductor MCU also requires low ESR ceramic capacitors to be configured near the analog power pins for decoupling and energy storage.

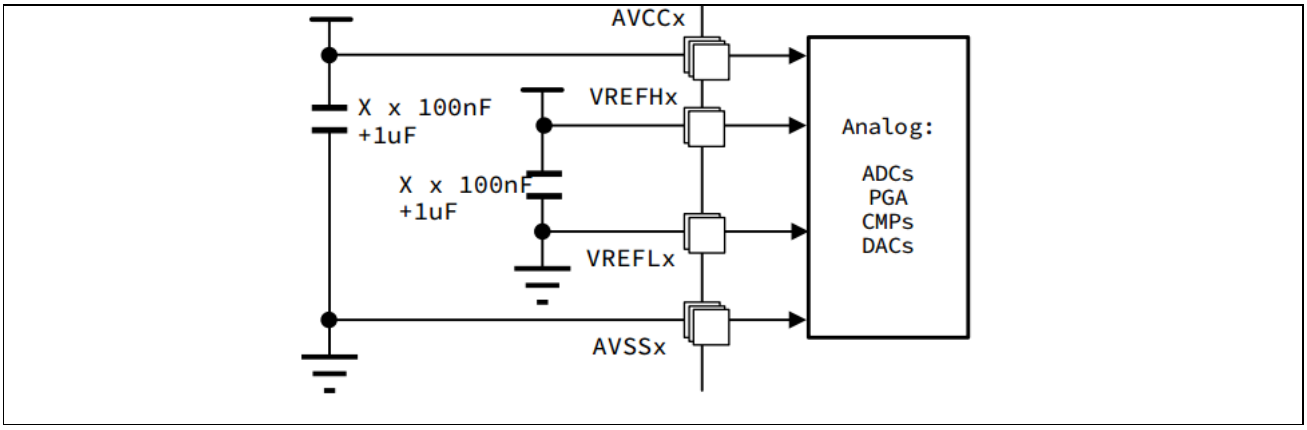


Figure 3-2 HC32F4A0 analog power capacitor

- When designing the PCB layout, ensure that the analog power loop is as small as possible and separate it from the digital ground, using single-point grounding. Place the energy storage and decoupling capacitors as close to the power pins as possible.

3.3 ADC input impedance and sampling time

In the ADC section of Xiaohua Semiconductor's MCU specification, the important parameters of the ADC model are generally listed, as shown in the following table:

表 3-51 ADC 特性

符号	参数	条件	最小值	典型值	最大值	单位
V_{AVCC}	电源	-	1.8	-	3.6	V
V_{REFH}(1)	正参考电压	-	1.8	-	V_{AVCC}	V
f_{ADC}	ADC 转换时钟频率	高速工作模式下 V_{AVCC}=2.4 ~3.6V	1	-	60	MHz
		低速工作模式下 V_{AVCC}=1.8 ~2.4V	1	-	30	
		超低速工作模式	1	-	8	
V_{AIN}	转换电压范围	-	V_{REFL}	-	V_{REFH}	V
R_{AIN}	外部输入阻抗	详见公式1	-	-	50	kΩ
R_{ADC}	采样开关电阻	-	-	3	6	kΩ
C_{ADC}	内部采样和保持电容	-	-	4	7	Pf
t_D	触发器转换延迟	f_{ADC} = 60 MHz	-	-	0.3	µs

The ADC sampling switch resistance RADC and the ADC internal sampling and holding capacitance CADC are indicated. The maximum external input impedance allowed is calculated according to the formula. The formula mentioned here is:

$$R_{AIN} \leq \frac{t}{C_{ADC} * \ln 2^{N+1}} - R_{ADC}$$

Where:

t: sampling period

N: ADC bit

The sampling circuit of SAR ADC can be simplified as shown below:

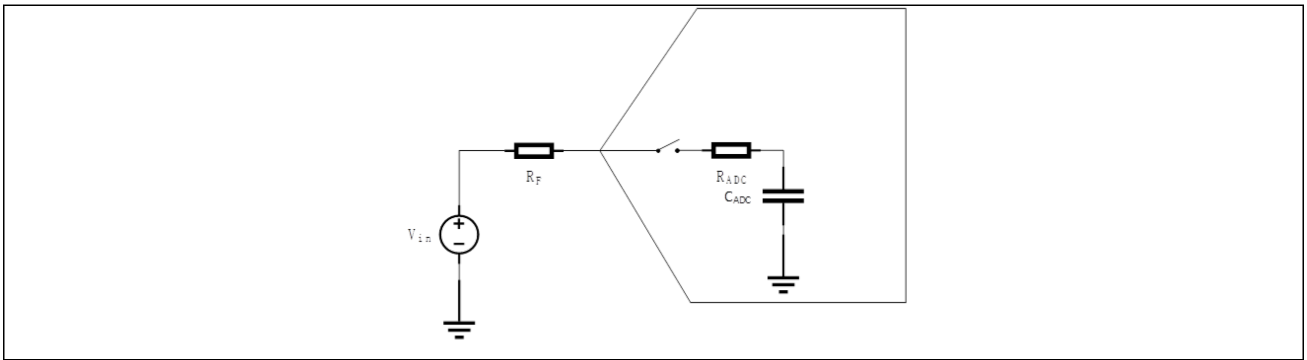


Figure 3-3 SAR ADC principle block diagram

When the ADC internal channel is switched to an external pin, the signal voltage on the pin will charge Cs through RAIN and RADC. Its charging time constant is:

$$\tau = (R_F + R_{ADC}) * C_{ADC}$$

Assuming that the initial voltage of Cs is V_{t0}, the system unit step response is:

$$V_t=V_{t0}+(V_{in}-V_{t0})*(1-e^{-\frac{t}{\tau}})$$

After a fixed time (i.e. sampling time t), the voltage difference between the voltage across the CADC capacitor and the signal source is:

$$V_{err}= (V_{in}-V_{t0})*e^{-\frac{t}{\tau}} \qquad (2)$$

When Vt0=0, that is, when the initial state is zero, the error is the largest. The larger Vin is, the larger the error is, and it is the largest when it is equal to VREF.

The charging curve is as follows:

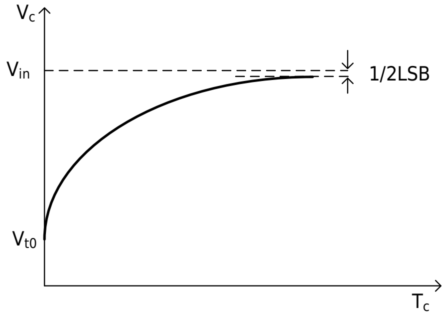


Figure 3-4 RC unit step response

If the sampling accuracy is required to reach 1/2 LSB, the error needs to meet the following requirements:

$$V_{in}*e^{-\frac{t}{(R_F+R_{ADC})*C_{ADC}}}\leq\frac{1}{2}*\frac{V_{ref}}{2^N}$$

Simplified, we can get:

$$R_F\leq \frac{t}{C_{ADC}*\ln 2^{N+1}}-R_{ADC} \qquad (3)$$

或

$$t\geq(R_F + R_{ADC}) *C_{ADC}*\ln 2^{N+1} \qquad (4)$$

Therefore, in order to obtain a higher sampling rate, it is necessary to reduce the external input impedance. If the external impedance is large, it is necessary to consider adding a driver stage for impedance conversion or increasing the sampling time.

In actual applications, the ADC input pin may be connected to an external RC filter circuit, as shown in Figure 3-5.

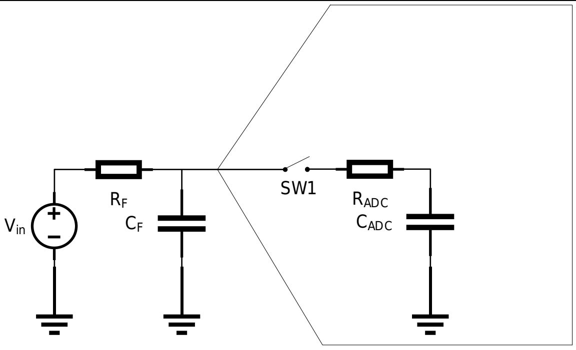


Figure 3-5 ADC external RC filter circuit diagram

In this case, charging is divided into 2 stages, and the process is shown in the figure below.

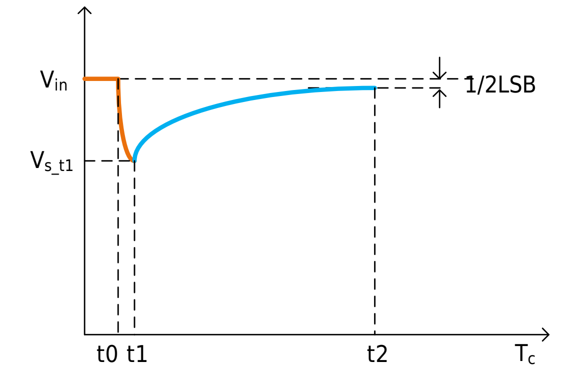


Figure 3-6 Capacitor balancing and charging process

Assume that the voltage across CF has reached the input signal amplitude Vin before the ADC channel switches to the external pin. After the internal switch is closed, the charges of CF and CADC are balanced, and finally the voltage balance is achieved.

The voltage after balance:

$$V_{s\_t1} = \frac{V_{in} * C_F}{C_F + C_{ADC}} \qquad (5)$$

At this equilibrium voltage, the input signal charges CF and CADC again through RAIN + RADC, and the starting point of charging is the voltage after equilibrium.

The time constant is:

$$\tau = (R_F + R_{ADC}) * (C_F + C_{ADC})$$

Substituting equation (5) into equation (2) yields:

$$V_{err} = \left(\frac{V_{in} * C_{ADC}}{C_F + C_{ADC}}\right) * (1 - e^{\frac{t}{\tau}})$$

If you want to achieve 1/2LSB accuracy, you must:

$$\left(\frac{V_{in} * C_{ADC}}{C_F + C_{ADC}}\right) * \left(1 - e^{-\frac{t}{(R_F + R_{ADC}) * (C_F + C_{ADC})}}\right) \leq \frac{1}{2} * \frac{V_{ref}}{2^N}$$

After simplification, we can get:

$$R_F \leq \frac{t}{(C_F + C_{ADC}) * \ln \frac{C_{ADC}}{C_F + C_{ADC}} 2^{N+1}} - R_{ADC} \qquad (6)$$

When $C_F = 0$, equation (6) is consistent with equation (3).

Here, the values of R_F and C_F also need to consider their bandwidth. Usually, in order to ensure that the input signal is not distorted after passing through the filter, the following conditions are met:

$$\tau_{AIN} \geq 10 * \tau_{FILT}$$

Right now:

$$f_{FILT} \geq 10 * f_{AIN} \qquad (7)$$

The 3dB bandwidth of the RC filter circuit is:

$$f_{3db}=\frac{1}{2\pi}f_{FILT} \qquad (8)$$

In addition, in actual systems, the initial voltage of Cs is often the voltage of the previous channel, so if the input voltages of two consecutively sampled channels differ greatly, the sampling accuracy of the first channel on the second channel may be affected.

Optimization suggestions:

- Ensure sufficient sampling time.
- Reduce external input impedance. For high-impedance input sources, it is necessary to increase the drive level.
- Add RC filtering to suppress input signal noise, and the RC bandwidth needs to be considered.
- Avoid arranging two signals with large voltage differences to adjacent sampling channels as much as possible.

3.4 Input signal

According to equation (1), it is not difficult to find that in addition to the reference source, another factor directly related to the conversion result is the input signal itself. When the input signal is small, the input signal noise has a significant impact.

If the impedance is too large, resulting in insufficient output current capacity, the sampling capacitor cannot be fully charged during high-speed sampling.

For AC signals, it is also necessary to pay attention to whether the signal has spike pulses or ripples. If an amplifier or RC filter circuit is used at the front end of the ADC. The bandwidth of the amplifier and the bandwidth of the RC need to ensure that the input signal is not distorted.

Optimization suggestions:

- Reduce the signal source impedance to provide sufficient charging current.
- Add a suitable RC filter to reduce the input signal noise. The RC bandwidth can be roughly estimated according to equations (7) and (8), or it can be accurately calculated according to the following formula:

$$\tau_{FILT\_MAX}=\frac{t}{\ln\left(\frac{2^{N+1}*2\pi f_{AIN}*V_{peak}*t_{CONV}*C_{ADC}}{V_{REF}*(C_F+C_{ADC})}\right)}$$

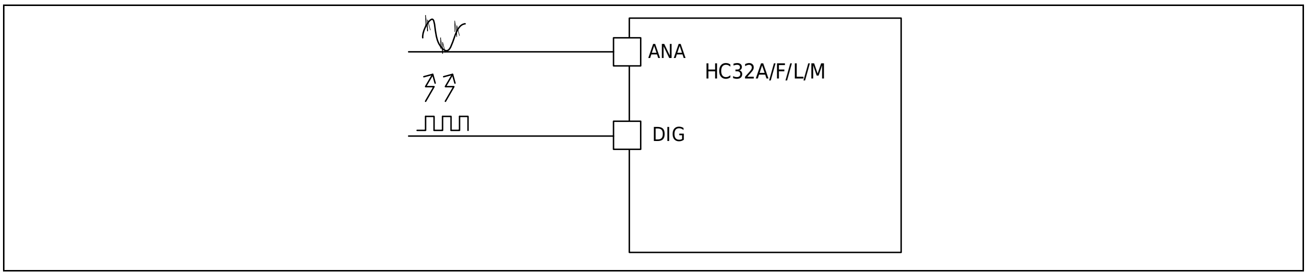
  The 3 dB bandwidth of the RC filter circuit is:

$$f_{3dB}=\frac{1}{2\pi*\tau_{FILT\_MAX}}$$

- Add an amplifier for impedance conversion.
- Trigger ADC sampling through accurate timing to avoid noise points of the input signal.

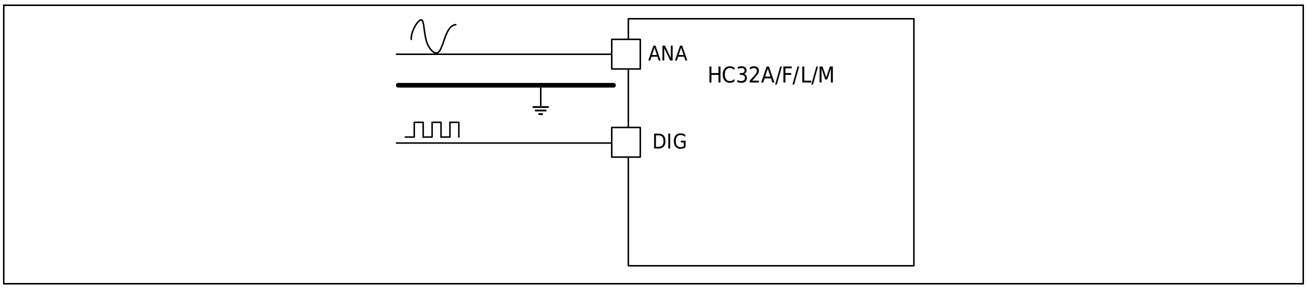
3.5 IO crosstalk

IO crosstalk may come from interference from adjacent lines on the circuit board, or from coupling inside the MCU port. This is especially true when there is a continuous high-speed square wave signal (communication signal, PWM signal, etc.) in close proximity to the ADC line. This square wave signal will generate spikes on the adjacent ADC line through capacitive coupling, affecting the ADC sampling accuracy.



Optimization suggestions:

- When arranging pin functions, try to avoid PWM, high-speed communication ports and ADC pins being adjacent.
- In PCB layout, keep ADC input signal lines away from PWM or high-speed communication lines.
- Use ground wires to isolate ADC input signal lines from PWM or high-speed communication lines.



3.6 Improving accuracy by software

Improving ADC sampling accuracy by optimizing power supply noise and external circuits is a fundamental means. However, in actual applications, the system electrical environment is often very complex, hardware improvements will also increase costs to a certain extent, and ADC inherent errors and white noise cannot be improved by hardware. Therefore, improving sampling accuracy by software is also an important means.

Optimization suggestions:

- Software filtering, according to noise characteristics, real-time, software overhead and other factors, use appropriate software filtering methods, such as average filtering, median filtering, first-order filtering, IIR and other filtering methods.
- Oversampling, oversampling can effectively improve the SNR of ADC and increase the effective bits of ADC data. It is suitable for occasions where the requirements for time real-time are not high and the MCU storage resources are relatively rich.

4 Summary

Successive approximation ADC (SAR ADC) has the advantages of fast speed, high accuracy and low cost, and is widely used in various MCUs. This application note briefly describes the working principle and inherent error of SAR ADC. It focuses on the factors that have a great impact on ADC sampling accuracy and how to optimize the design in practical applications. For example: reducing system noise, controlling input impedance, extending sampling time, etc. In addition, in some applications, it is also a common way to reasonably use other peripherals to trigger ADC sampling to avoid noise.

In short, ADC, as the last link in the signal chain, is crucial to the stability, accuracy and real-time performance of control. In addition to being related to the ADC itself, the power supply and external circuit design also need to match the ADC. Xiaohua Semiconductor is committed to providing high-performance MCUs, and also provides customers with strong technical support to reduce the burden on R&D personnel and shorten the time to market.

Revision history

Version	Date	Revision Content
Rev1.0	2023/11/08	Initial version released.